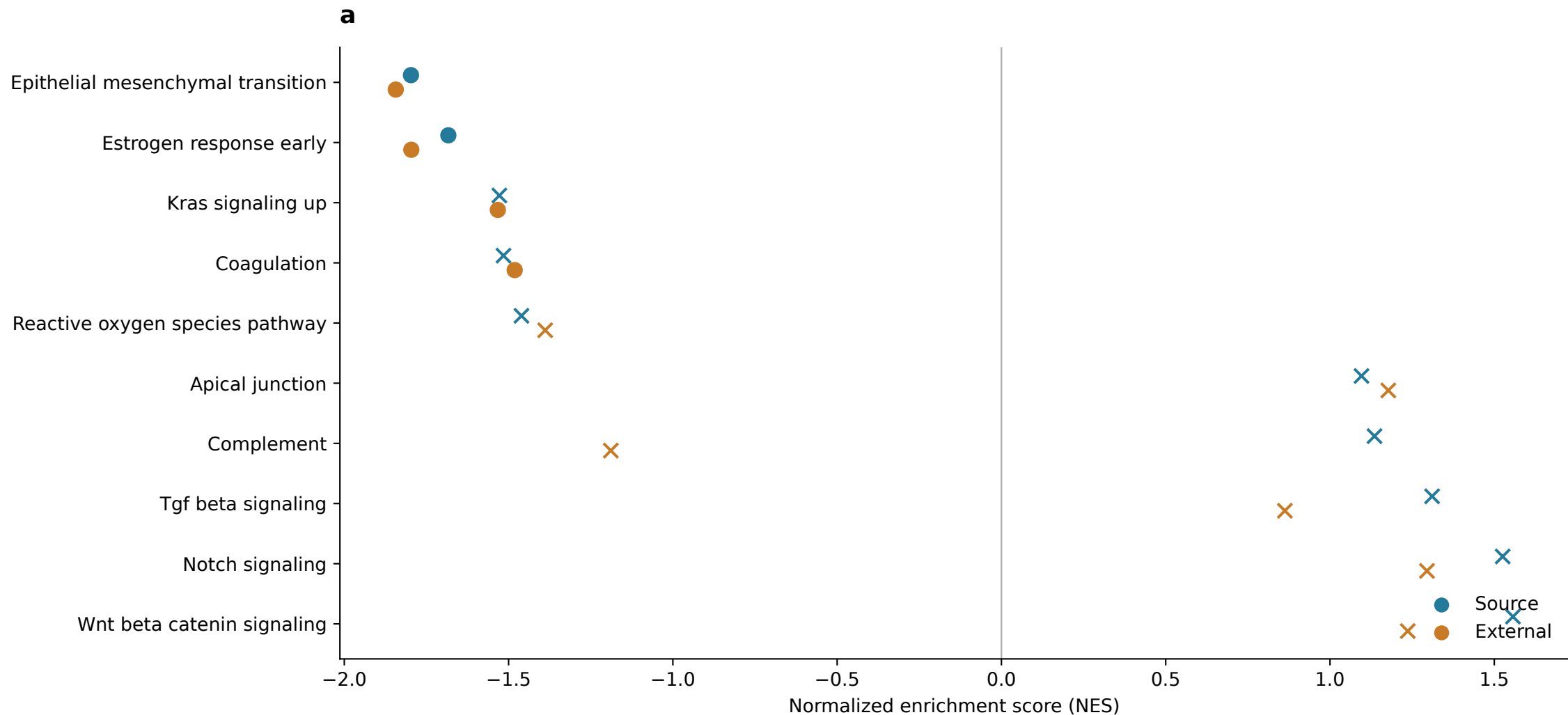


Hallmark GSEA of genome-wide fibroblast contrasts



Five strongest positive and negative source NES, with external estimates retained. Circles: gene-set FDR ≤ 0.05 ; crosses: > 0.05 . 1,000 gene-set permutations; these q-values are not donor-level biological significance tests. All 50 sets per cohort are tabulated.